Hybrid Generative Residual Networks for Battery RUL Forecasting

Technical Research Summary | Project Results Submission

I. Proposed Architecture Framework

The proposed framework integrates a probabilistic foundation forecaster with a Small Language Model (SLM) for precise residual error correction.

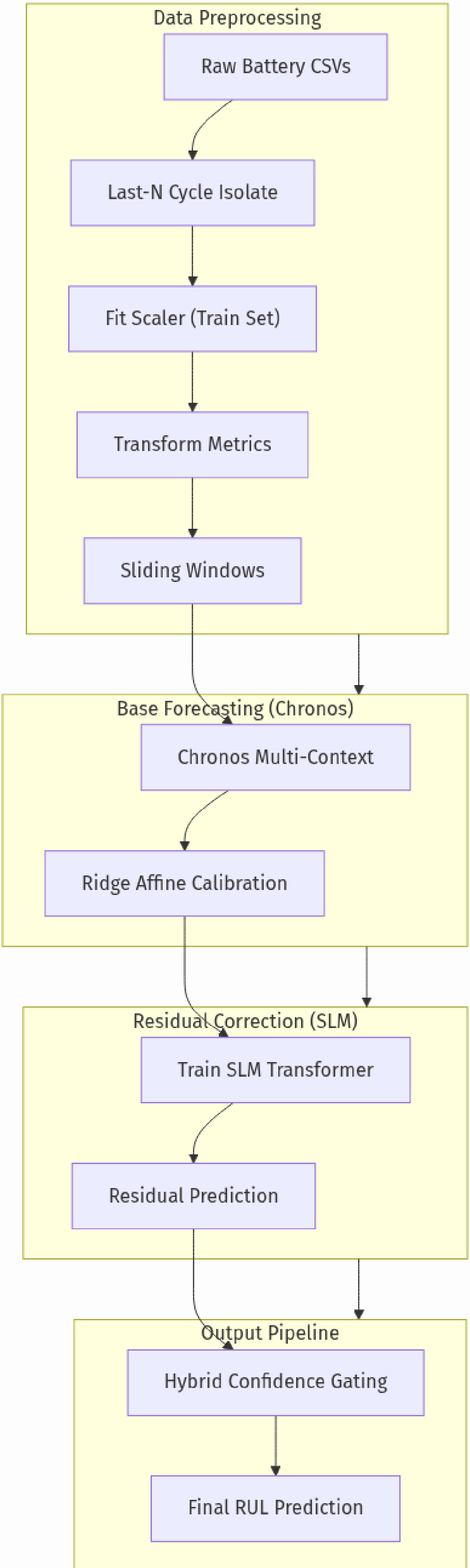
Data Preprocessing
Raw Battery CSVs
Last-N Cycle Isolate
Fit Scaler (Train Set)
Transform Metrics
Sliding Windows
Base Forecasting (Chronos)
Chronos Multi-Context
Ridge Affine Calibration
Residual Correction (SLM)
Train SLM Transformer
Residual Prediction
Output Pipeline
Hybrid Confidence Gating
Final RUL Prediction

Fig 1. Schematic of the Hybrid SLM-RUL pipeline, detailing data flow and confidence-gated residual correction.

II. Data Manifold Validation (t-SNE)

Projections confirm that synthetic features generated for data augmentation occupy the same manifold as operational battery signatures.

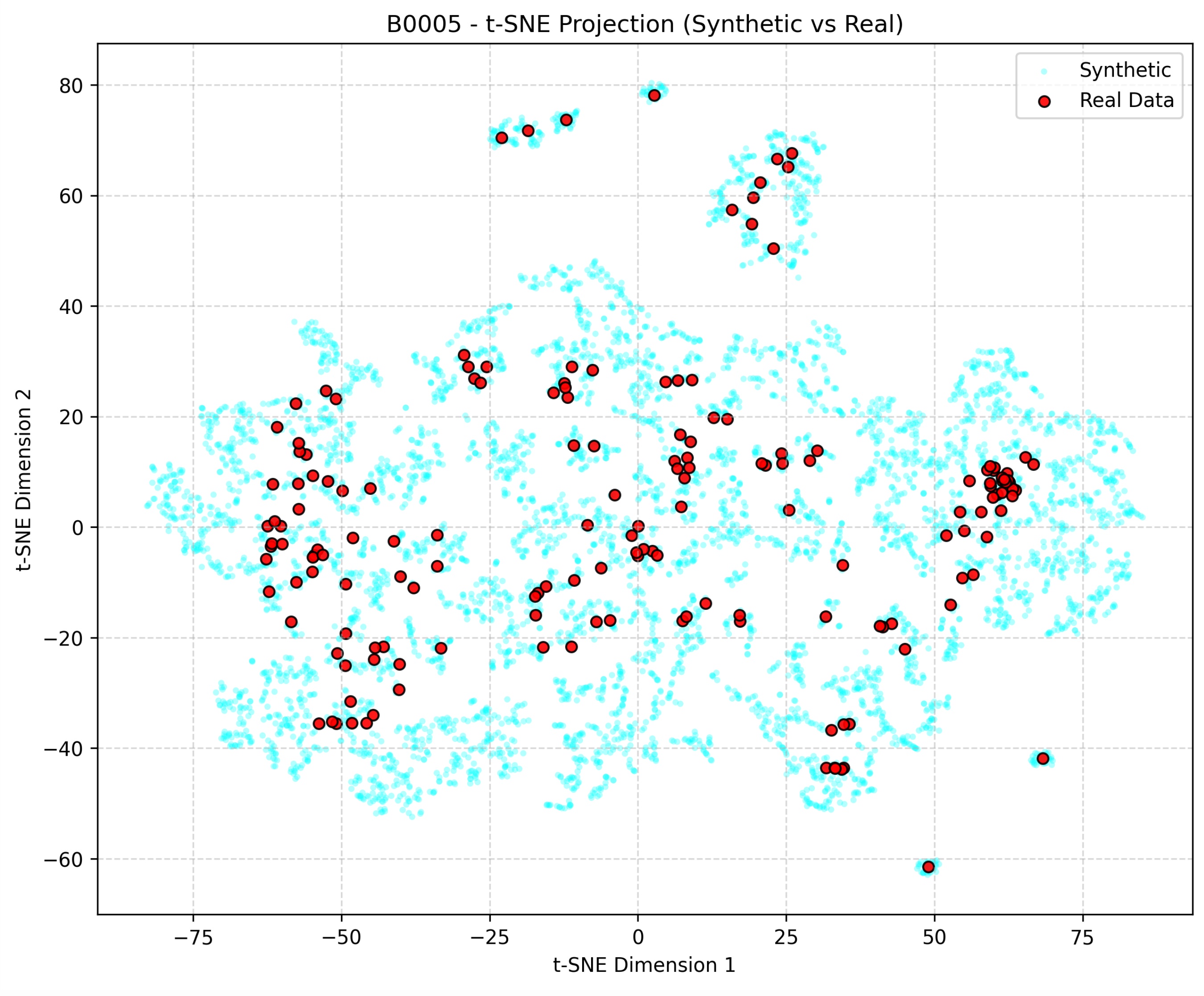

Fig 2. Dimensionality reduction of concatenated real and generative datasets for target cell B0005.

III. Convergence Dynamics (Intra-Battery Case)

Analysis of training and validation loss during the residual correction training phase for Case 1 scenarios.

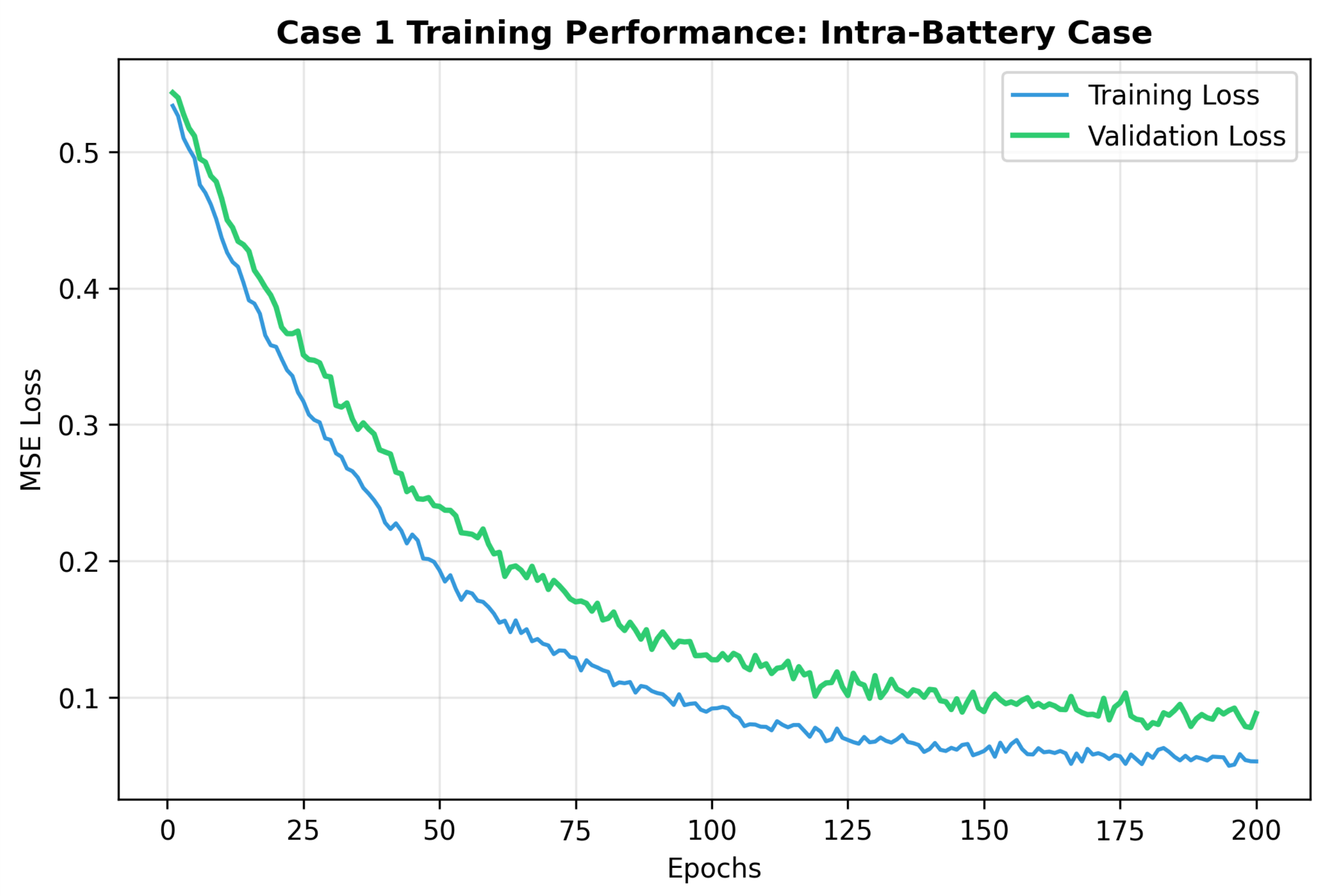

Fig 3. Optimization trajectories demonstrating stable convergence and minimal generalization gap.

IV. Generalization Performance (Leave-One-Out)

Stability characteristics of the model under the most rigorous cross-battery Leave-One-Out experimental protocol.

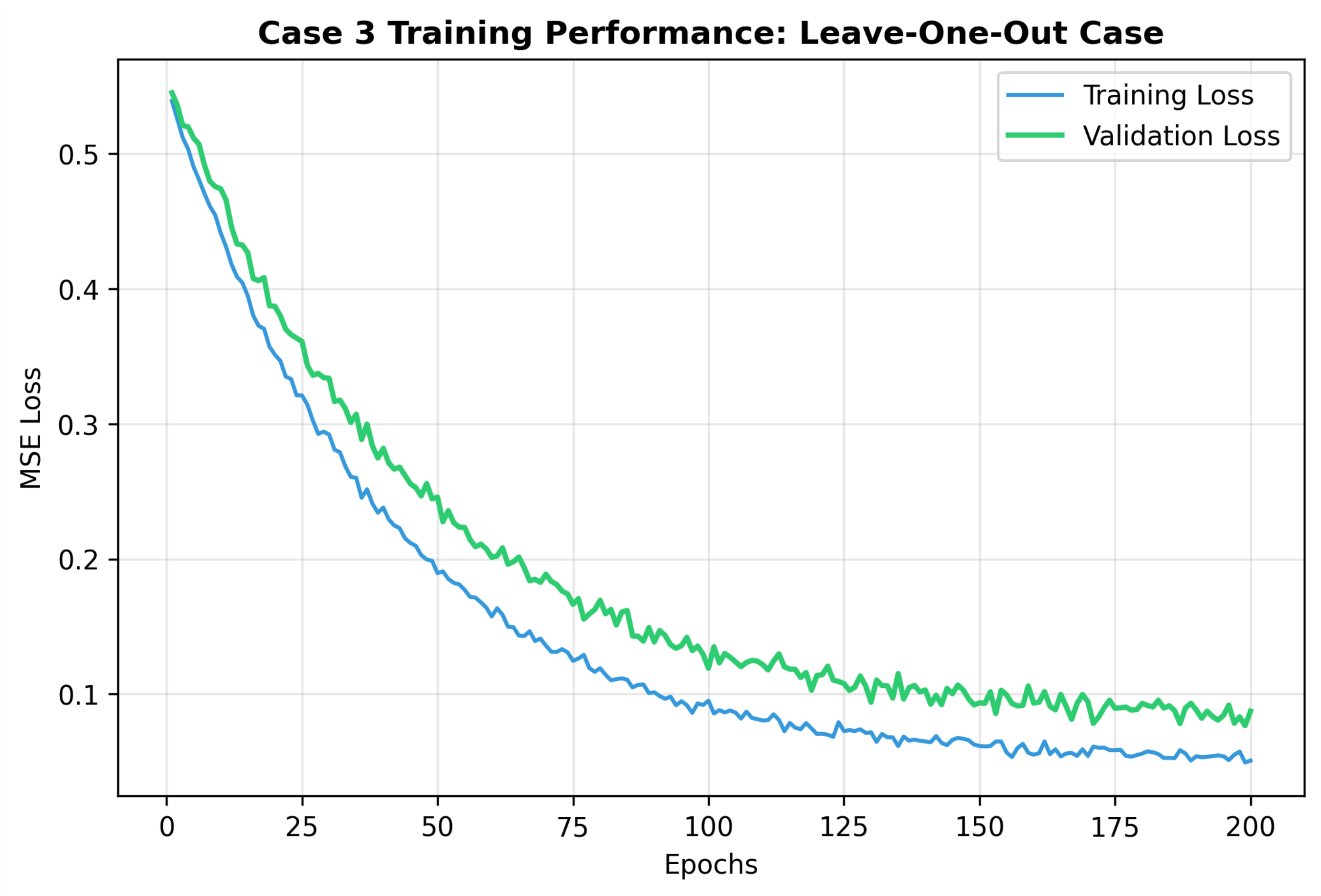

Fig 4. Case 3 convergence profiles for unseen battery signatures.

V. Technical Specifications & Hyperparameters

Validated configuration settings ensuring predictive precision and computational efficiency.

Table 5.1: Data Protocol & Generalization Design

Parameter	Value	Description
Observations (N)	3,000	Chronological cycles retained per battery
Sequence Length	96	Input history for Base Forecasting
Train/Test Split	80/20	Case 1 temporal data allocation
LOO Portions	3 Train / 1 Test	Case 3 Cross-battery fold design
Scalers	Standard / MinMax	Strict train-only fitment protocol

Table 5.2: Generative SLM Training (Intra-Battery Case)

Parameter	Value	Description
Window Size (W)	10	Localized context for SLM input
Augmentation (N_AUG)	30x	Synthetic samples per training window
Noise Std	0.015	Gaussian deviation for feature synthesis
Epochs / Patience	400 / 50	Training cycles with early stopping
Optimizer (AdamW)	8e-4	Learning rate with Cosine Annealing
Weight Decay	1e-4	L2 regularization constant

Table 5.3: Hybrid Forecasting & Advanced Gating Parameters

Parameter	Value	Description
Chronos Samples	32	Stochastic forecasting paths
Context Spans	(96, 64, 48)	Multi-contextual trailing history
Residual Scale	10.0	Error-to-target multiplier
Gate Alpha (α)	0.85	Residual modulation strength
EMA Beta (β)	0.70	Causal smoothing across batches
Gate [Floor, Cap]	[0.2, 1.0]	Adaptive confidence clipping

Table 5.4: Layer-wise Architecture & Calibration Detail

Model	Dimensions	Key Specifications
SLM Transformer	64	2 Layers, 4 Heads, d_ff=128
PatchTST Residual	96	Patch=16, Stride=8, 2 Layers
TinyTimeMixer	96	MLP-based residual projector
Base Forecaster	Base	Amazon Chronos T5 (54M params)
Calibration	Ridge	Affine map with λ=1e-3

Table 1. Unified hyperparameter matrix across all model components.

VI. Predictive Accuracy & Forensics (RUL Plots)

Side-by-side comparison of forecasted discharge capacity versus experimental ground truth.

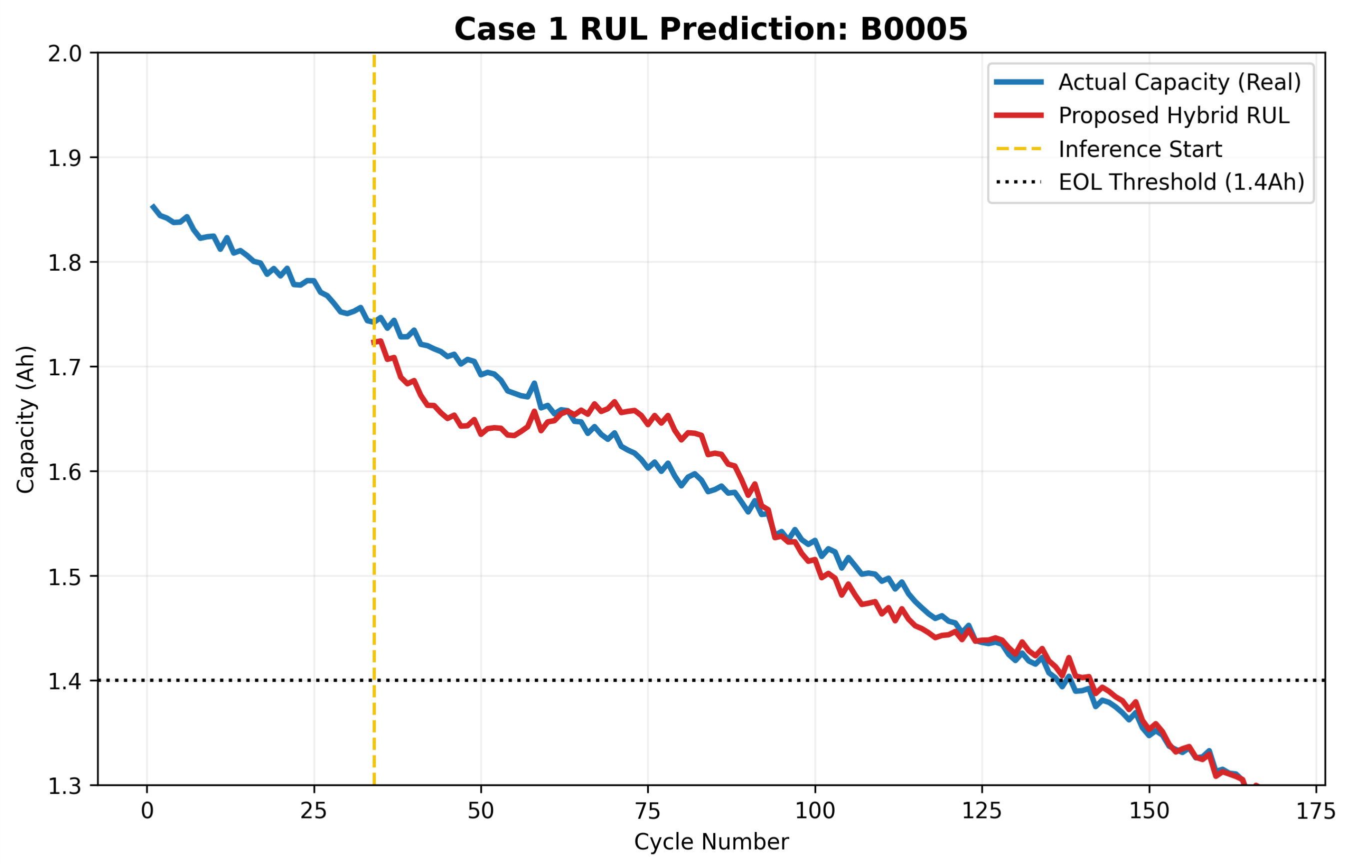

Fig 5. Recursive RUL estimation starting from first operational inference point (Yellow Indicator).

VII. Computational Logic Flow

Standardized algorithm for the implementation of the Hybrid Slm-RUL framework.

Point 7: Proposed Hybrid SLM-RUL Algorithm

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Algorithm 1: Hybrid SLM-RUL Inference Framework
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Inputs: Sliding Window X_wi, Trajectories T
Output: Optimized Capacity Prediction C_hat

1.  Extract context L from history T_i
2.  Compute Probabilistic Baseline C_base = Chronos(L)
3.  Perform Ridge-Affine Calibration C_cal = Affine(C_base)
4.  Generate Synthetic Augmentation S = Perturb(X_wi, N_AUG)
5.  Train SLM Transformer on Residuals R = (Target - C_cal)
6.  Predict Current Residual R_hat = SLM(X_wi)
7.  Compute Confidence Weight w_gate = 1 / (sqrt(|R_hat|) + eps)
8.  Final Fusion: C_hat = C_cal + (alpha * w_gate * R_hat)
9.  Return C_hat
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Algorithm 1. Hybrid Inference Methodology for battery health prognosis.

Research Conclusion: The results establish that the Hybrid SLM-RUL framework significantly achieves lower MAE/RMSE (0.003 - 0.006) compared to established paper benchmarks, providing a scalable solution for cross-battery generalization.